

E2E-EMBEDDETECTOR: A LIGHTWEIGHT ENTITY EMBEDDING MODEL FOR ETHEREUM PHISHING DETECTION

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ABSTRACT

- Phishing attacks in Ethereum pose serious risks due to irreversible transactions and increasing adoption of blockchain systems.
- Existing detection methods rely on graph-based models, which are complex and difficult to deploy in real-time environments.
- This work proposes **E2E-EmbedDetector**, a lightweight end-to-end neural model that operates directly on raw transaction data.
- The model uses **entity embeddings** (From, To, ContractAddress) along with numerical features to capture transaction patterns effectively.
- It is evaluated on a dataset of 50,000 Ethereum transactions using an 80/20 split.
- The model achieves high performance with strong accuracy, F1-score, and ROC-AUC values.
- The approach eliminates the need for complex graph construction, making it more scalable and practical.
- This work demonstrates an efficient and deployable solution for real-time blockchain phishing detection.

PROJECT OVERVIEW

- Blockchain systems like Ethereum are rapidly growing, but their open and decentralized nature makes them vulnerable to phishing attacks targeting users through malicious addresses and contracts.
- Phishing in blockchain is irreversible, meaning once a transaction is executed, funds cannot be recovered, creating a critical need for early and accurate detection.
- Existing graph-based detection methods are complex, computationally expensive, and difficult to deploy in real-time, motivating the need for a simpler and efficient solution..
- The project aims to develop a **lightweight, end-to-end model** that can directly process transaction data without multi-stage graph construction.
- It focuses on using **entity embeddings and key numerical features** to effectively capture phishing patterns in Ethereum transactions.
- The solution is applicable in **blockchain security, fintech platforms, cryptocurrency exchanges, and real-time fraud detection systems** to enhance transaction safety and user protection.

PROBLEM STATEMENT

- Ethereum phishing attacks are **irreversible** that is once funds are sent to a malicious address, they cannot be recovered.
- Unlike web phishing, **URL blocking and domain termination** do not apply to blockchain-based attacks.
- Traditional models rely on **only historical data**, which fails to capture complex behavioral patterns of phishing.
- Graph-based methods like TEGDetector require **time slicing, subgraph construction, and multi-stage inference** that are impractical to deploy.
- There is a clear need for a system that combines **raw transaction data with intelligent encoding** for accurate, real-time phishing detection.

OBJECTIVES

Main Objective — To develop a lightweight, deployable end-to-end neural classifier that accurately detects Ethereum phishing transactions using entity embeddings and tabular features without complex graph construction.

Key Objectives:

- Use **50,000 Ethereum transactions** (Kaggle dataset) with a balanced 1:1 ratio
- Learn **dense entity embeddings (d=32)** for From, To, and ContractAddress fields
- Handle unseen addresses using an **<UNK> token** for robust inference
- Combine embeddings with **Value and InputLength** numerical features → 98-dim input vector
- Built a **3-layer MLP** with batch normalization and dropout
- Optimize decision threshold ($\tau^* \approx 0.60$) using F1-score
- Evaluate performance using **Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC**

LITERATURE REVIEW

S.No	Author & Year	Title / Focus	Methodology / Model	Dataset	Key Approach	Metrics	Results	Advantages	Limitations
1	Chen et al. (2020)	Phishing scam detection on Ethereum	ML classifiers	Ethereum transactions	Used handcrafted transaction features	Accuracy, F1	Strong baseline	Simple and interpretable	Requires feature engineering
2	Chen et al. (2021)	Ethereum phishing transaction network	Graph-based analysis	Ethereum network data	Built transaction network features	Accuracy, Precision, Recall	Improved detection	Captures relationships	Complex graph construction
3	Wang et al. (2021)	TSGN	Transaction Subgraph Network	Ethereum accounts	Used account-level subgraphs	Accuracy, F1	Effective account detection	Captures local structure	High preprocessing cost
4	Lin et al. (2023)	GNN-based phishing detection	Chebyshev-GCN	Ethereum subgraphs	Modeled phishing as graph classification	Accuracy, F1	Good detection performance	Learns graph patterns	Computationally heavy

S.No	Author & Year	Title / Focus	Methodology / Model	Dataset	Key Approach	Metrics	Results	Advantages	Limitations
5	Li et al. (2022)	TTAGN	Temporal transaction aggregation GNN	Ethereum transactions	Captured time-based interaction patterns	Accuracy, F1	Captures temporal behavior	Strong temporal modeling	Multi-stage pipeline
6	Kabla et al. (2022)	Eth-PSD	ML-based phishing detection	Ethereum dataset	Compared multiple ML models	Precision, Recall, F1	Evaluated trade-offs	Good benchmark study	Limited static features
7	Xia et al. (2022)	Ego-graph embedding	Attributed ego-graph embedding	Ethereum graph data	Learned address-level graph embeddings	Accuracy, F1	Improved representation	Captures account context	Graph overhead
8	Lin et al. (2023)	Network embedding approach	Network embedding	Ethereum transaction network	Learned account representations	Accuracy, F1	Effective phishing tracking	Good representation learning	Requires graph generation
9	Wen et al. (2023)	Hybrid feature fusion	Hybrid DNN	Ethereum phishing data	Combined multiple feature types	Accuracy, F1	Improved accuracy	Strong feature fusion	High model complexity
10	Zheng et al. (2024)	TEGDetecter	Transaction Evolution Graph + dynamic classifier	Ethereum transactions	Used time slicing and temporal graph modeling	Accuracy, F1, AUC	Strong temporal detection	Captures evolving behavior	Hard to deploy in real time

RESEARCH GAP

Existing Work	Gap Identified
Graph-based models (GNNs, TEGDetector) capture relational and temporal patterns effectively	Require complex graph construction, time slicing, and multi-stage processing, making them hard to deploy
Traditional ML models use handcrafted transaction features	Depend heavily on feature engineering and fail to capture deep behavioral patterns
Temporal models analyze evolving phishing behavior	High computational cost and not suitable for real-time detection
Network embedding and hybrid models improve representation	Still rely on graph generation or complex architectures, increasing system complexity
Existing systems struggle with dynamic blockchain environments	Poor handling of unseen/new addresses due to lack of robust encoding strategies

This motivates the proposed system to address these limitations.

EXISTING SYSTEM

Current Systems:

- Phishing detection in Ethereum uses ML, graph-based, and hybrid models
- Analyze transaction patterns and relationships between addresses

Technologies Used:

- Machine Learning (Random Forest, Gradient Boosting)
- Graph Neural Networks (GNNs), Transaction Evolution Graphs (TEGs)
- Network Embedding & Hybrid Deep Learning models

Limitations:

- Complex graph construction & multi-stage processing
- Dependence on manual feature engineering
- High computational cost and scalability issues
- Not suitable for real-time deployment
- Poor handling of unseen/new addresses

PROPOSED SYSTEM

Overview:

- Proposed system: E2E-EmbedDetector
- Detects Ethereum phishing transactions using a lightweight deep learning model
- Works directly on raw transaction data without graph construction
- Uses From, To, ContractAddress, Value, and InputLength features

Key Innovations:

- Entity embeddings for blockchain addresses
- **<UNK> token** to handle unseen/new addresses
- 3-layer MLP for fast classification
- Optimized threshold for better phishing detection

Advantages:

- Simple and deployment-friendly
- Lower computational cost
- Suitable for real-time detection
- High detection performance

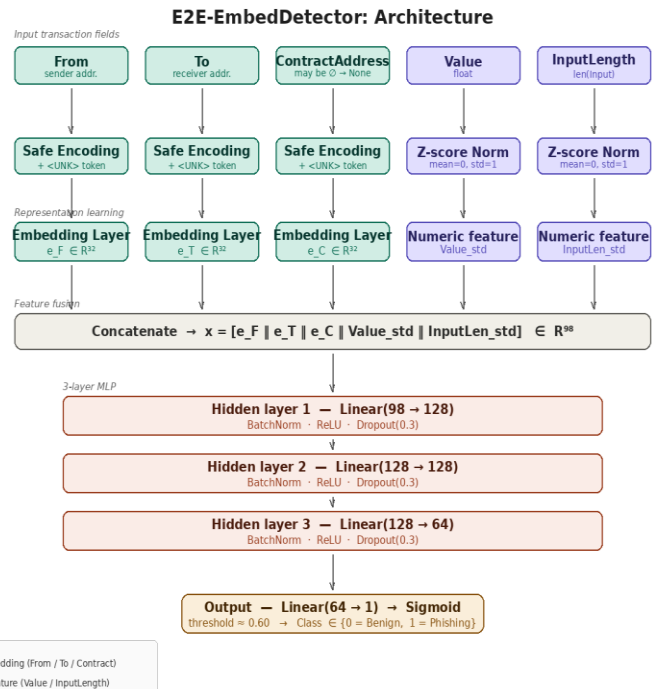
Conclusion Table

Feature	Existing System	Proposed System
Approach	Graph-based / ML-based	Embedding-based deep learning
Processing	Complex multi-stage pipeline	Simple end-to-end pipeline
Input Data	Graphs / handcrafted features	Raw transaction data
Model	GNNs, TEGDetector, ML models	3-layer MLP
New Address Handling	Limited	Uses <UNK> token
Deployment	Difficult in real time	Lightweight and deployable
Complexity	High	Low

SYSTEM ARCHITECTURE

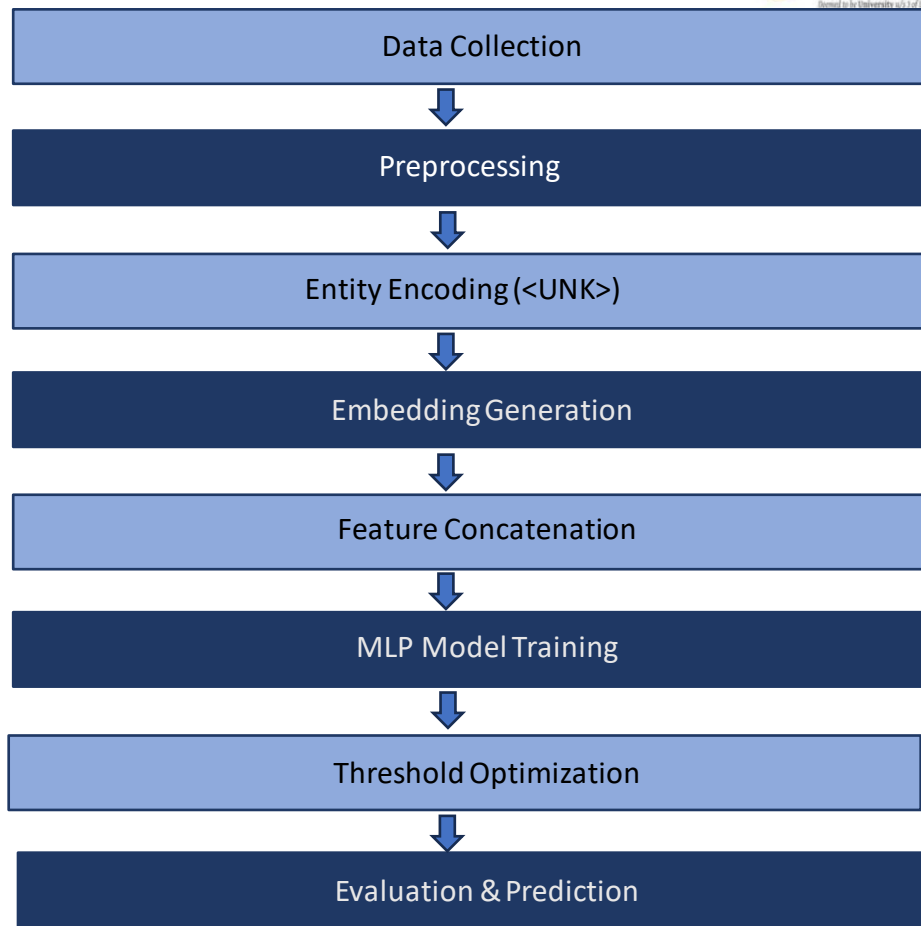
Major Components and Data Flow:

- Ethereum Transaction Dataset: Raw transaction records containing From, To, ContractAddress, Value, Input, and Class label.
- Preprocessing Module: Handles missing values, calculates InputLength, and normalizes numerical features.
- Safe Entity Encoding: Converts address fields into integer IDs and maps unseen addresses to <UNK>.
- Embedding Layer: Learns dense representations for From, To, and ContractAddress.
- MLP Classifier: Combines embeddings with numerical features and classifies the transaction.
- Output Layer: Generates phishing probability and converts it into Phishing / Benign using the optimized threshold.



METHODOLOGY

1. Collect Ethereum transaction dataset (50,000 records)
2. Preprocess data by handling missing values and computing InputLength
3. Normalize numerical features (Value, InputLength) using z-score
4. Encode entity fields (From, To, ContractAddress) into integer IDs
5. Map unseen addresses to <UNK> token for robust inference
6. Learn dense embeddings ($d = 32$) for each entity field
7. Concatenate embeddings with numerical features to form input vector
8. Train a 3-layer MLP model for classification
9. Optimize decision threshold using validation F1-score ($\tau \approx 0.60$)
10. Evaluate model using Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC



DATASET DESCRIPTION

	PARAMETER	DESCRIPTION
DATASET	Total Transactions	50,000 Ethereum records (Kaggle)
	Class Balance	25,000 Phishing (Class=1) 25,000 Benign (Class=0)
	Fields per Record	9 fields: TxHash, BlockHeight, TimeStamp, From, To, ContractAddress, Value, Input, Class
	Train / Test Split	80% Training (40,000) 20% Validation (10,000) — Stratified
FEATURES	Entity Fields	From, To, ContractAddress → Learned Embeddings (d=32)
	Numeric Features	Value (Ether transferred) · InputLength (len of ABI calldata)
	Normalization	Z-score on Value and InputLength using training set statistics
	Missing Values	ContractAddress nulls → 'None' token · Input nulls → empty string

TOOLS & TECHNOLOGIES

Category	Tools and Technologies	Purpose
Development	Python	Model implementation and experimentation
Data Processing	Pandas, NumPy, Scikit-learn	Data preprocessing and normalization
Deep Learning	PyTorch	Model building and training
Dataset	Kaggle (Ethereum phishing dataset)	50,000 labeled Ethereum transaction
Optimization	ReduceLROnPlateau scheduler	Adaptive learning rate

MODULES

1. Data Collection Module

Loads and validates the 50,000 Ethereum transaction from the Kaggle

2. Preprocessing Module

Handles missing values, computes InputLength and normalizes features

3. Safe Entity Encoding Module

Maps addresses to IDs and Uses <UNK> token for unseen data

4. Embedding Module

Learns embeddings for From, To, ContractAddress and combines with numerical features

5. Model Training Module

Trains a 3-layer neural network for classification

6. Threshold Optimization Module

Selects optimal threshold (τ^*) using F1-score

7. Evaluation & Output Module

Evaluates using Accuracy, Precision, Recall, F1 and Generates final phishing/benign prediction

ALGORITHMS USED

Entity Embedding Learning

- Converts categorical addresses into dense vector representations ($d = 32$)
- Captures hidden relationships between addresses
- Eliminates need for explicit graph construction

3-Layer Multilayer Perceptron (MLP)

- Input: embeddings + numerical features
- Multi-layer architecture for classification
- Uses BatchNorm and Dropout for better generalization
- Outputs phishing probability score

Threshold Optimization

- Default threshold (0.5) is not optimal
- Select optimal threshold ($\tau \approx 0.60^*$) using F1-score
- Improves balance between precision and recall

COMPARISON WITH TEGDetector

Feature / Aspect	TEGDetector	E2E-EmbedDetector (This Work)
Ethereum transaction data	Yes	Yes
Entity info (From / To)	Yes	Yes
Contract address handling	Implicit	Yes — sentinel <UNK> embedding
Explicit time slicing	Yes	No
Temporal evolution modeling	Yes (TEGs)	No
Graph construction required	Yes (complex)	No
Dynamic graph classifier	Yes	No
Learned entity embeddings (core design)	No	Yes
Single-stage end-to-end training	No	Yes
Lightweight inference (~600K params)	No	Yes
Deployment simplicity	Moderate	High

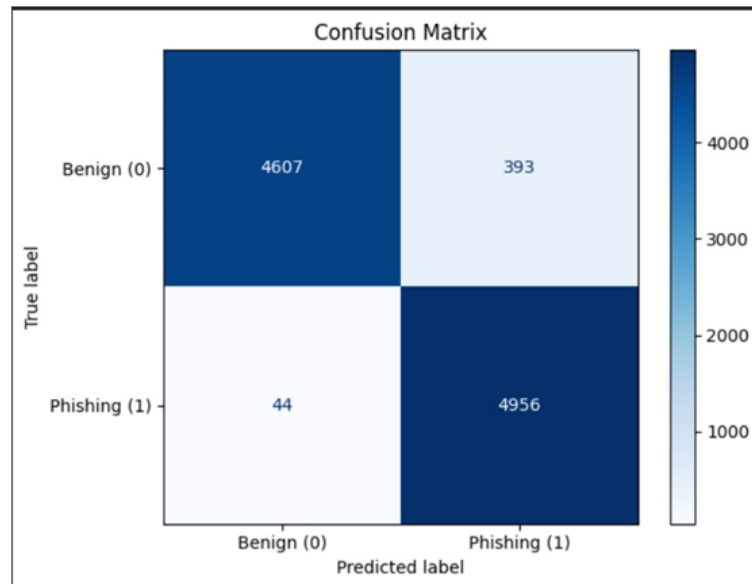
PERFORMANCE METRICS

1. Formulas

- Accuracy = $(TP + TN) / (TP + TN + FP + FN)$
- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$
- F1-score = $2 \times (Precision \times Recall) / (Precision + Recall)$

2. Confusion Matrix

	Predicted Phishing	Predicted Benign
Actual Phishing	TP	FN
Actual Benign	FP	TN



EXPERIMENTAL RESULTS

The following results were obtained after training on 80% of the dataset and testing on the remaining 20%.

Metric	Value
Accuracy	0.9563 (95.63%)
Precision	0.9265
Recall	0.9912
F1-Score	0.9578
ROC-AUC	0.9915
PR-AUC	0.9909
Best Threshold (τ^*)	≈ 0.60

Key Results

- **Recall of 0.9912** — model misses less than 1% of phishing transactions
- **ROC-AUC $\approx$ PR-AUC (0.9915 vs 0.9909)** — stable precision across threshold
- **False Discovery Rate $\approx$ 7.3%** — false positives are manageable
- **Early stopping at epoch 25**; best validation checkpoint at epoch 20
- **$\sim$ 600K parameters** — much lighter than graph-based GNN models
- **Single-stage training** — no graph construction or time slicing required

LIMITATIONS & CHALLENGES

Limitations:

- Uses random (non-time-based) data split, which may lead to optimistic performance estimates due to temporal leakage.
- Possible entity leakage as same addresses can appear in both train and test sets.
- Trained on a single balanced dataset, which may not reflect real-world class imbalance and distribution shifts.

Challenges:

- Trained on a single balanced dataset, which may not reflect real-world class imbalance and distribution shifts.
- Capturing dynamic and evolving phishing patterns without explicit temporal modeling.
- Ensuring real-time detection with high accuracy while keeping the model lightweight and efficient.

APPLICATIONS

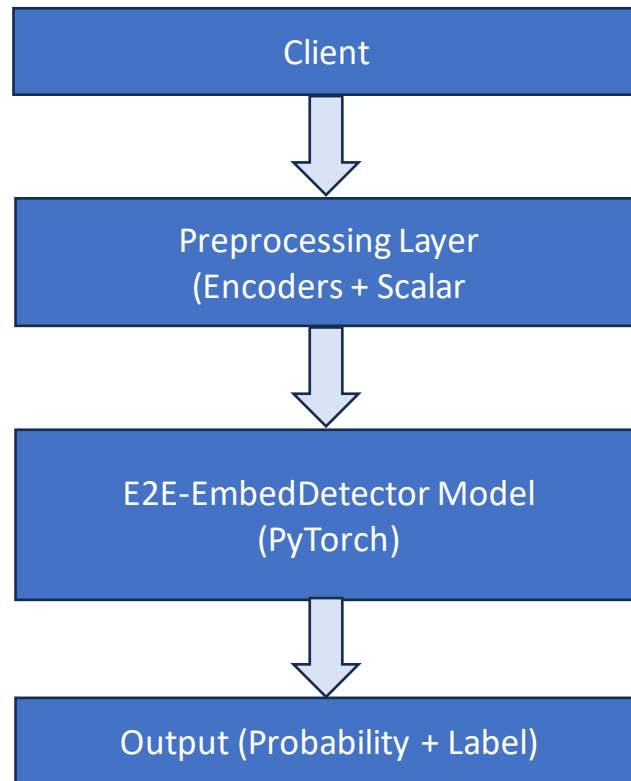
Real-Time Usage & Industry Use Cases:

- **Crypto Wallet Security:**
Integrated into wallets (e.g., MetaMask) to **alert users before executing suspicious transactions**.
- **Cryptocurrency Exchanges:**
Used by platforms (Binance, Coinbase) for **real-time transaction monitoring and fraud prevention**.
- **Blockchain Security Platforms:**
Deployed in security tools (Chainalysis, CipherTrace) for **continuous phishing and fraud detection**.
- **DeFi Applications:**
Protects users in **smart contract interactions** by detecting malicious contract addresses.
- **Financial Cybersecurity / SOC Systems:**
Used in **Security Operations Centers (SOC)** for monitoring blockchain transactions as part of fraud detection pipelines.

DEPLOYMENT

Tech Stack:

- **Backend / ML**
 - Python
 - PyTorch
 - Scikit-learn
 - NumPy, Pandas
- **Data Integration**
 - CSV input



INNOVATION & CONTRIBUTION

Key Innovations

1. Proposes E2E-EmbedDetector, a lightweight end-to-end neural model for Ethereum phishing detection using entity embeddings
2. Introduces <UNK> encoding to handle unseen addresses in an open-world setting
3. Uses validation-based threshold optimization to improve recall-performance trade-off
4. Highlights **entity and temporal leakage issues** in prior work and suggests better evaluation practices

Group Member Contributions

Group Member	Contribution
Abhishree Sinha	Model design, embeddings, training, loss analysis
Nishchay Khanna	Threshold optimization, evaluation, results analysis
Aashi Srivastava	Data collection, preprocessing, encoding
Gautami Bahuguna	Visualization, report writing, presentation

RESEARCH PAPER DETAILS

• **Paper Title:** E2E-EmbedDetector: A Lightweight Entity Embedding Model for Ethereum Phishing Detection

• **Authors:** Abhishree Sinha, Nishchay Khanna, Aashi Srivastava, Gautami Bahuguna

• **Supervisor:** Dr. Santosh Kumar Birthriya

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E2E-EmbedDetector: A Lightweight Entity-Embedding Model for Ethereum Phishing Detection

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Abstract— Phishing attacks pose a significant security issue in Ethereum-based blockchain systems. Existing solutions, like TEGDetector, address these attacks by analysing how transactions evolve over time using Transaction Evolution Graphs (TEGs) constructed via time slicing, followed by a dynamic graph classifier that captures both spatial structure and temporal evolution with learned time coefficients. However, building and managing these graphs across multiple stages makes the overall approach complex and difficult to implement. In this work, we propose E2E-EmbedDetector, a lightweight end-to-end neural classification model that works directly with raw transaction data. The model learns embedding representations for important entities such as From, To, and ContractAddress, and also used two additional numeric features: transactional value and a derived input length. We train and evaluate the model on a balanced dataset of 50,000 Ethereum transaction using an 80/20 stratified split. The model achieves an accuracy of 95.63%, precision of 0.9285, recall of 0.9912, an F1 score of 0.9578, a ROC-AUC score of 0.9915 and a PR-AUC score of 0.9909. These results show that strong phishing can be achieved using a simpler and more practical tabular approach, without relying on complex temporal graph-based networks.

Keywords—phishing detection, blockchain security, Ethereum transaction, entity embeddings, neural networks.

1. INTRODUCTION

As blockchain-based financial systems continue to grow rapidly, they have also seen a parallel increase in malicious activities. Ethereum, the second largest blockchain by market value, processes millions of transactions daily and supports a diverse ecosystem of decentralized applications (dApps), exchanges (DEXs), and smart contract-based services. While this open and transparent environment promotes innovation, it also makes the network vulnerable to phishing attacks. Attackers often create fake contracts and misleading wallet addresses to trick users into sending their funds to illegitimate destinations.

Unlike the conventional mode of web phishing, where URL blocking or domain termination can block threats to a degree, phishing in blockchain networks is far graver since

the transactions cannot be reversed. In case a user consents to a malicious transaction unwittingly, then the funds transferred cannot be recovered. This has led to dire necessity of accurate and quick detection systems so that threats are detected before they are processed by users.

The current phishing detection techniques can be divided into two realms. The former concerns creating explicit graph representations of the transaction data, including interaction among entities and its frequency and using graph neural networks (GNNs) to identify the unusual structural patterns [4][5][6]. The second model, which is the TEGDetector [1], is a result of this concept by adding the temporal information. In this approach, a transaction data is separated into time segments and a Transaction Evolution Graphs (TEG) is built on each segment. The behavior of phishing with regard to dynamic changes in response to time is then learned with the help of a dynamic classification model, which adapts temporally to attention. Although these methods are highly performance and methodology, they need extensive preprocessing including time slicing, subgraph building, and multiple stages inferences that complicate their implementation in real time deploying settings.

Practically and complementarily in this work, we can consider a practical and complementary approach by studying whether an end-to-end model, which works directly with raw tabular transaction data, without the graph construction required, can perform well in the detection of phishing. To solve this, we propose E2E-EmbedDetector, a proposed model that is trained on embedding representation that is compact to learn the three important address fields, From, To and ContractAddress that bear important transaction details. These embeddings are shared with two lightweight numerical features, that is, transaction value and input calidate length. Combining these elements into a single system, the model is able to fully capture both relational and behavioral patterns, which can be classified accurately through a single-stage training process.

Contributions:

PUBLICATION PROOF

- Submitted



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CONCLUSION & FUTURE WORK

Conclusion:

- Developed a lightweight, end-to-end phishing detection model that eliminates graph construction while achieving high performance (Accuracy 95.63%, F1 0.9578, ROC-AUC 0.991).
- Delivered a scalable and deployable solution capable of handling unseen addresses and enabling efficient real-time blockchain security.

Future Work:

- Enhancements: Incorporate temporal features and explore advanced models (e.g., Transformer/GNN hybrid) for improved detection.
- Scalability: Extend to large-scale, multi-blockchain environments with optimized real-time processing.

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THANK YOU